

Oussama Jalleli

☎ +1 (873) 456-3175 | ✉ oussama.jalleli@inrs.ca | in LinkedIn | GitHub | 📍 Gatineau, QC, Canada

RESEARCH INTERESTS

Multimodal signal processing and audio-visual perception modelling; trustworthy and explainable AI (XAI) for media forensics; biosignal-driven human-computer interaction; cross-modal representation learning; deep learning for speech, audio, and video understanding; robustness and generalization in perception systems; self-supervised and transfer learning for signal processing.

PUBLICATIONS & PRESENTATIONS

O. Jalleli, Y. Zhu, and T. Falk, “Leveraging Speech-to-Facial-Electromyography Mapping for Singing Voice Deepfake Source Tracing: A Pilot Experiment,” *ECCV*, 2026.

O. Jalleli, Y. Zhu, and T. H. Falk, “Audio-Visual Cross-Attention for Improved Deepfake Video Detection and Forgery Localization,” *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, Vienna, Austria, 2025. DOI: 10.1109/SMC58881.2025.11342953

RESEARCH EXPERIENCE

Graduate Researcher — MuSAE Lab, UMR INRS-UQO

Gatineau, QC, Canada

Multimodal Signal Processing & Trustworthy AI

January 2025 – Present

- Developing **biosignal processing pipelines** (EMG) for **physiological consistency validation** in **multimodal perception systems**; enabling **trustworthy AI** by grounding predictions in human biology and improving **model interpretability**.
- Architecting **audio-visual cross-modal attention mechanisms** for **temporal segmentation** and **anomaly localization** in **multimedia signal processing**; improving **forensic traceability** and **model robustness** across diverse **adversarial manipulation** scenarios.
- Implementing **self-supervised learning** and **transfer learning** strategies to enhance **generalization** across unseen manipulation types and datasets.

AI Research Intern — MuSAE Lab, UMR INRS-UQO

Gatineau, QC, Canada

Multimodal Signal Processing & MLOps

May 2024 – August 2024

- Engineered a pipeline for **multimodal signal processing**: raw media ingestion → **feature engineering** → **model inference** → decision output.
- Leveraged **scene context modeling** and **attention-based feature extraction** to improve **model robustness** and **generalization** across diverse **adversarial manipulation** scenarios.
- Built **real-time inference** capabilities for joint audio/video forensic analysis.

Research Collaborator — Behavioural Neuroscience Lab, Lund University

Lund, Sweden

Computer Vision & Biomechanical Signal Analysis

November 2023 – August 2024

- Developed a **computer vision pipeline** for **3D pose estimation** and **biomechanical signal analysis**; automated **behavioral phenotyping** for neuroscience research.
- Applied **deep learning-based keypoint detection** (CNNs) to enable automated **time-series analysis** of gait patterns in mouse models.

AI Engineering Intern — IP-Label

Tunis, Tunisia

LLM-Powered Report Automation & Production ML

June 2023 – August 2023

- Built a **production LLM application** for **automated report generation** from **time-series telemetry data**; deployed via **REST API**, reducing report turnaround from hours to minutes and integrating into company **analytics workflow**.
- Designed **feature engineering** pipelines for raw telemetry ingestion, enabling **real-time anomaly detection** and automated insight extraction.
- Implemented **CI/CD** with **Jenkins** and **model monitoring** dashboards using **Prometheus** and **Grafana** to track **model drift** and performance degradation.

Mentorship & Leadership

Mitacs Globalink Mentor	Remote / Gatineau, QC, Canada
<i>Part-Time Research Mentor — Undergraduate Research Internships</i>	<i>April 2026 – Present</i>
- Mentoring international undergraduate researchers in their 12-week research internships at Canadian universities. - Organizing social integration events and providing ongoing weekly support for cultural adaptation, campus orientation, and professional development in STEM research environments.	

Education

Institut National de la Recherche Scientifique (INRS)	Montréal, QC, Canada
<i>M.Sc. in Telecommunications</i>	<i>January 2025 – Present</i>
- Thesis: EMG-based Deepfake Detection and Facial Activity Mapping — multimodal signal processing, biosignal analysis, trustworthy AI - Supervisor: Prof. Tiago H. Falk — MuSAE Lab	
Esprit School of Engineering	Ariana, Tunisia
<i>B.Eng. in Data Science Engineering</i>	<i>September 2021 – September 2024</i>
Higher Institute of Applied Sciences and Technology (ISSAT)	Sousse, Tunisia
<i>Integrated Preparatory Cycle (Mathematics & Physics)</i>	<i>September 2019 – June 2021</i>

Technical Skills

Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras; CNNs, Transformers, GANs, GNNs/GATs, YOLO, LLMs; self-supervised learning , transfer learning , adversarial robustness
Signal & Media Processing: OpenCV, librosa, FFmpeg; audio-visual feature extraction , biosignal processing (EMG), speech processing , time-series analysis , 3D pose estimation
MLOps & Production: Docker, Jenkins, Git; model deployment , CI/CD , model monitoring (Prometheus, Grafana), experiment tracking , REST API development
Programming: Python (NumPy, Pandas, Scikit-learn), C/C++, MATLAB
Data & Infrastructure: SQL, Oracle; Linux (Ubuntu); UML High-Performance Computing: SLURM, MPI, CUDA, multi-GPU training, distributed computing, cluster job scheduling

Certifications

Applications of AI for Anomaly Detection	November 2023
<i>NVIDIA Deep Learning Institute Verify Certificate</i>	
Building Transformer-Based NLP Applications	October 2023
<i>NVIDIA Deep Learning Institute Verify Certificate</i>	
Natural Language Processing Specialization	November 2023
<i>DeepLearning.AI (Coursera)</i>	

Languages

Arabic: Native	French: Bilingual (C2)	English: Fluent (C2)	German: Beginner (A1)
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Interests

Guitar Playing		Cinematography		Football		Reading (Science, Philosophy)
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